

AIKAWA, Japan

WMO: 476020

Lat: 38.0292N

Lon: 138.2400E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	-0.4	-7.6	2.0	1.9	-6.7	2.1	2.2	19.4	2.0	18.1	2.2	9.0	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	31.5	25.0	30.2	24.8	29.1	24.5	26.1	29.5	25.6	29.0	25.0	28.4	3.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	28.5	24.6	19.6	28.0	23.9	18.8	27.5	81.2	29.7	78.9	29.0	76.3	28.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.6	12.8	-2.7	34.4	1.0	1.7	-3.4	35.6	-4.0	36.6	-4.6	37.6	-5.4	38.8		
(5)			-4.0	26.9	0.8	0.7	-4.6	27.4	-5.0	27.8	-5.5	28.1	-6.1	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	3.8	3.7	6.2	11.0	16.0	19.8	24.2	26.1	22.5	17.1	11.6	6.7	(6)
(7)		DBStd	8.17	2.42	2.68	3.04	3.31	2.99	2.30	2.55	2.19	2.75	2.74	3.12	2.95	(7)
(8)		HDD10.0	649	193	177	124	26	1	0	0	0	0	0	19	109	(8)
(9)		HDD18.3	2175	451	409	376	221	85	11	0	0	2	56	203	361	(9)
(10)		CDD10.0	2152	1	1	6	56	186	294	439	499	375	221	67	7	(10)
(11)		CDD18.3	634	0	0	0	1	12	56	181	241	126	18	0	0	(11)
(12)		CDH23.3	4316	0	0	0	2	30	118	1200	2235	700	31	0	0	(12)
(13)		CDH26.7	1083	0	0	0	0	2	9	282	647	141	2	0	0	(13)
(14)	Wind	WSAvg	4.9	8.1	6.9	5.7	4.3	3.4	3.0	2.8	3.1	3.6	4.6	6.0	7.8	(14)
(15)	Precipitation	PrecAvg	1602	138	96	96	93	101	127	192	139	139	134	164	183	(15)
(16)		PrecMax	2059	273	196	181	166	234	237	476	399	348	232	284	385	(16)
(17)		PrecMin	1067	57	36	46	18	46	49	19	7	37	80	96	91	(17)
(18)		PrecStd	238	42	35	31	41	52	63	116	85	77	42	40	61	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	14.6	17.5	22.7	26.1	27.9	33.1	34.0	32.0	26.0	20.5	16.4	(19)
(20)			MCWB	9.2	9.7	10.4	14.4	17.2	21.0	24.6	25.1	24.8	20.3	15.6	12.6	(20)
(21)		2%	DB	10.4	11.5	14.7	20.1	24.0	25.9	30.7	31.9	29.8	23.9	19.1	14.4	(21)
(22)			MCWB	7.4	7.9	8.9	13.0	16.6	20.4	24.4	25.1	23.8	19.0	15.1	10.9	(22)
(23)		5%	DB	9.1	9.7	12.9	18.2	22.4	24.6	29.4	30.8	28.3	22.6	18.0	12.9	(23)
(24)			MCWB	6.2	6.8	8.8	12.1	16.1	19.9	24.4	25.4	23.4	18.2	14.2	9.7	(24)
(25)		10%	DB	7.8	8.2	11.2	16.4	20.9	23.5	28.2	29.8	26.8	21.6	16.8	11.6	(25)
(26)			MCWB	5.2	5.5	8.1	11.3	15.5	19.6	23.9	25.0	22.6	17.3	13.2	8.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	10.8	11.7	16.2	19.2	22.9	26.2	27.1	25.9	21.9	17.1	13.3	(27)
(28)			MCDWB	12.0	12.6	14.8	19.7	22.1	25.3	29.7	30.5	29.3	24.9	19.2	15.6	(28)
(29)		2%	WB	7.8	8.5	10.5	14.2	18.1	21.9	25.5	26.5	24.6	20.1	15.9	11.3	(29)
(30)			MCDWB	10.0	11.0	13.3	18.1	21.7	24.0	29.0	29.9	28.4	22.9	18.3	13.8	(30)
(31)		5%	WB	6.5	7.0	9.4	13.1	17.2	21.1	24.9	26.0	23.8	19.0	14.7	9.9	(31)
(32)			MCDWB	8.8	9.4	12.2	16.7	20.8	23.4	28.3	29.3	27.4	21.8	17.6	12.7	(32)
(33)		10%	WB	5.3	5.6	8.3	12.1	16.2	20.4	24.3	25.6	23.1	17.8	13.5	8.8	(33)
(34)			MCDWB	7.6	8.1	11.1	15.3	19.8	22.8	27.6	28.9	26.5	20.9	16.4	11.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.3	4.9	5.9	6.8	6.9	5.9	5.0	5.7	5.9	6.2	5.9	5.0	(35)
(36)			MCDBR	6.7	7.4	8.6	9.7	9.3	7.3	6.5	6.8	6.9	7.0	7.3	6.9	(36)
(37)			MCWBR	5.9	6.4	6.3	5.7	5.1	4.1	3.4	3.8	4.0	4.7	6.0	6.1	(37)
(38)		5% WB	MCDBR	6.5	7.1	7.5	8.3	7.9	5.6	5.6	5.8	6.1	6.1	6.9	6.8	(38)
(39)			MCWBR	6.0	6.6	6.3	5.5	5.1	3.6	3.3	3.6	3.9	4.8	6.4	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.425	0.458	0.517	0.551	0.556	0.555	0.566	0.529	0.475	0.446	0.436	0.433	(40)	
(41)		taud	2.029	1.951	1.845	1.817	1.874	1.960	1.998	2.112	2.214	2.159	2.091	2.011	(41)	
(42)		Ebn at Noon	708	741	738	744	750	749	737	755	775	751	694	662	(42)	
(43)		Edh at Noon	127	156	191	208	200	183	175	153	131	125	119	122	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.25	3.54	4.71	5.32	5.30	4.85	5.10	3.99	3.04	1.87	1.27	(44)	
(45)		RadStd	0.13	0.24	0.29	0.41	0.56	0.45	0.66	0.56	0.29	0.23	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page